



Short Communication

MOGAD optic neuritis after mild head/orbital trauma in six children

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ABSTRACT

Myelin oligodendrocyte glycoprotein antibody-associated disease (MOGAD) is a potential cause of optic neuritis (ON). Its triggers and etiologies are not completely understood. We describe a novel clinical observation in six young patients with MOGAD-ON in the setting of strikingly parallel histories of mild head/orbital trauma. This is a single-center retrospective case series of six young patients and age-matched isolated MOGAD-ON controls. We present data both individually (de-identified, only the six trauma-associated cases) and in aggregate. Averages are presented as the arithmetic mean $\pm$ SEM. 6/27 patients with MOGAD-ON, (3/6 female), between 8 and 18 years old presented with ON 5.5 days after mild head trauma. Four patients developed ON ipsilateral to their unilateral head trauma while two developed bilateral ON following midline head trauma. All patients tested positive for serum anti-MOG antibodies upon ON workup. They all received intravenous corticosteroids with rapid improvement in symptoms (5.5 weeks to full visual recovery) and none have since relapsed. No other patients with MOGAD-ON experienced preceding head trauma, and all patients in the control group were asked about trauma upon assessment of the history. Head trauma may serve as an inciting event in the presentation and diagnosis of MOGAD-ON. This novel observation provides a potential pathophysiologic mechanism independent of infectious triggers, although we cannot determine if these patients were already predisposed towards MOGAD-ON.

1. Introduction

Myelin oligodendrocyte glycoprotein antibody-associated disease (MOGAD) is an inflammatory demyelinating disease of the central nervous system characterized by antibodies (Ab) targeting MOG expressed on the surface of oligodendrocytes (Marignier et al., 2021; Adelman et al., 1995). MOGAD was previously thought to be an uncommon demyelinating disorder in children, but recent studies have illustrated increased diagnosis and awareness (de Mol et al., 2020). Diagnosis is dependent on the clinical presentation, radiographic evidence of CNS demyelination and serological evidence of anti-MOG antibodies in serum with the gold-standard cell-based assay (Banwell et al., 2023). Optic neuritis (ON) is a common clinical presentation of MOGAD, particularly in older children (Santoro et al., 2023), which can manifest as unilateral or bilateral eye pain, vision loss, optic disc swelling, and extensive perineural and optic head lesions on orbital MRI (Chen et al., 2018; Ciotti et al., 2022).

While the precipitant for MOGAD is unknown, infectious agents are widely thought to trigger the seroconversion into anti-MOG antibody production (Rusescu et al., 2017; Zaini et al., 2023; Kang et al., 2023), although disease-specific pathogens have not been isolated (Marignier et al., 2021). Severe orbital and periorbital trauma is known to cause optic neuropathy (Karimi et al., 2021), but MOGAD-ON secondary to or precipitated by blunt force has not previously been reported. Here, we present six children with strikingly similar histories and phenotypes who warrant reporting a potential etiology of MOGAD-ON following mild orbital/cranial trauma (Maas et al., 2017).

2. Methods

We identified six children in our inpatient and emergency room settings between 2016 and 2023 with no prior pertinent medical history who were diagnosed with MOGAD-ON, as determined by clinicoradiographic evidence of ON and the presence of anti-MOG antibodies

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detected in blood (all Mayo Labs, Rochester, MN and all with a titer of at least 1:100), following trauma to the head in the preceding 14 days. Examination findings, laboratory evaluation from serum and CSF (if applicable), imaging, and follow up outcomes were abstracted from electronic medical records. Snellen visual acuity was converted to

LogMAR to standardize arithmetic comparisons. We compared clinical characteristics of the six children with trauma-associated MOGAD-ON to a control group of 22 children with first-onset MOGAD-ON *not* associated with trauma between 2016 and 2023. Continuous numerical variables are expressed as arithmetic mean $\pm$ SEM.

Table 1
Clinical Characteristics of Each MOGAD-ON Case.

	Patient 1	Patient 2	Patient 3	Patient 4	Patient 5	Patient 6
Demographics						
Age at ON Onset (Sex)	18.1 (M)	14.2 (M)	10.5 (F)	8.3 (F)	12.5 (M)	7.8 (F)
Race/Ethnicity	White/Hispanic	White/Hispanic	White/Hispanic	White/Not Hispanic	Asian/Not Hispanic	White/Not Hispanic
Clinical Characteristics						
Head Trauma Description	Football struck above right orbit, no LOC	Head-on collision with peer while playing football, no LOC	Left temporal blunt trauma with sibling (head-on-head), no LOC	Struck back of head on corner of picture frame, no LOC	Basketball struck right anterior temporal area, no LOC	Sibling's hand struck left suborbital and nasal area, no LOC
Latency to ON Symptoms (days)	3	14	1	7	10	2
Affected Eye(s)	Right	Bilateral	Left	Bilateral	Right	Left
Associated with Pain	Yes	Yes	No	Yes	Yes	No
Latency to Presentation for ON (days)	8	7	14	1	3	1
Ophthalmologic Exam at ON Onset						
Visual Acuity (R,L)	20/200, 20/20	20/25, 20/200	20/20, hand motion	20/25, 20/200	20/50, 20/25	20/20, NLP
Ishihara Color Plate (R,L)	NA	11/11, 2/11	4/4, 0/4	NA	NA	NA
Subjective Color Desaturation	Yes	Yes	Yes	Unknown	Yes	Yes
Fundusoscopic Exam	mild nasal elevation	bilateral grade 2 disc edema	grade 2 disc edema	bilateral grade 1 disc edema	grade 1 disc edema	grade 2 disc edema
Imaging						
Orbit	Right orbital apex to intracranial enhancement	Bilateral intra-orbital optic nerve enhancement	Left intra-orbital, intracranial and distal cisternal segments enhancement	bilateral intra-orbital and left intracranial enhancement/enlargement	Right intra-orbital enhancement/enlargement	Left intra-orbital to cisternal enhancement/enlargement
Brain	Normal	Normal	Normal	Normal	Normal	<i>ex vacuo</i> dilatation of supratentorial ventricular system
Spine	Normal	NA	Normal	Normal	NA	Normal
Infectious Workup						
EBV Serology Status at Presentation	Remote Infection	NA	Inconclusive	NA	Negative	Negative
Syphilis Screening	Negative	Negative	Negative	NA	Negative	Equivocal
TB Screening	NA	Negative	Negative	NA	Negative	NA
Lyme Screening	Negative	NA	Negative	NA	NA	NA
ON Serological Workup						
Anti-MOG Antibody	1:1000	1:1000	1:1000	1:100	1:100	1:1000
Anti-AQP4 Antibody	Negative	Negative	Negative	Negative	Negative	Negative
FANA at Presentation	Negative	NA	Positive (1:320)	NA	NA	Negative
Anticardiolipin IgG/IgM	Negative	NA	Negative	NA	NA	NA
Lysozyme	1.89	2.02	1.96	NA	0.75	0.63
CSF ACE	1.2	NA	0.9	NA	46	Negative
Treatments						
IVMP (30 mg/kg/day 3–5 days, Max 1 g/day)	Yes	Yes	Yes	Yes	Yes	Yes
IVIG	No	Yes	Yes	No	No	No
PLEX	No	No	Yes	No	No	No
Oral Prednisone	60 mg, 12 days	60 mg, 42 days	30 mg, 42 days	1 mg/kg, 15 days	60 mg, 18 days	45 mg, 56 days
Taper (starting dose, duration)						
Outcomes						
Latency to Best Post-ON Visual Acuity	9 weeks	6 weeks	7 weeks	5 weeks	4 weeks	3 weeks
Best Post-ON Visual Acuity	20/20, 20/20	20/20, 20/20	20/20, 20/20	20/20, 20/20	20/20, 20/20	20/20, 20/20
Relapsed (Y/N, Duration of Follow Up)	No, 10 months	No, 11 months	No, 18 months	No, 60 months	No, 1 month	No, 26 months

ON, optic neuritis; LOC, loss of consciousness; IVMP, intravenous methylprednisolone; IVIG, intravenous immunoglobulin; PLEX, plasma exchange;

2.1. Standard protocol approvals, registrations, and patient consents

The study was approved by the Institutional Review Board at Baylor College of Medicine (Houston, TX) and each of the six patients were consented for publication.

3. Results

Six children presented with trauma-associated MOGAD-ON (age 13.5 $\pm$ 1.8 vs 11.0 $\pm$ 0.7 years, 3/6 female vs 10/21 female, 3/6 Hispanic vs 13/21 Hispanic in the control MOGAD-ON group) and are presented in Table 1. The head trauma was significant enough for the patients or their families to reference it with the front-line provider prior to being asked about history of trauma. The median latency between trauma and the onset of ON symptoms was 5 days (range 1–14 days) and there were neither preceding visual or other symptoms (such as headache, eye pain, or color desaturation) prior to the trauma nor during the interim period after the trauma and before the onset of ON symptoms. The description of head trauma was qualitatively similar among the six patients: two patients were struck in the periorbital area by a ball while participating in an athletic activity (1 basketball, 1 football, without loss of consciousness but severe enough to come out of the activity for the rest of the day), one child suffered a head-on collision with a peer while playing football, two were struck by a sibling during a “play-fight”, and one child inadvertently struck the back of their head against the corner of a sharp picture frame hung on the wall. Two cases of ON were bilateral and four were unilateral (two left-sided). All the unilateral ON cases were ipsilateral to the trauma, and the bilateral cases were all preceded by midline trauma. MRI brain and orbits with and without gadolinium revealed optic nerve enhancement and swelling in a pattern consistent with MOGAD-ON in all six patients without the presence or stigmata of head trauma and anti-MOG antibody was positive in serum 5.5 days (range 1–17 days) after ON symptom onset and 14.5 days (range 6–21 days) after the inciting trauma. All six patients were treated with high dose intravenous methylprednisolone (IVMP, 30 mg/kg/day or max dose of 1000 mg per day) for 3–5 days with at least some subjective improvement in symptoms, two additionally received intravenous immunoglobulin (IVIG), and one of those patients also received plasmapheresis with subsequent considerable improvement. None of the six patients have had ON or other CNS demyelinating relapses with an average of 14.5 months (range 1–60 months) since ON. Visual acuity improved to normal in all six patients within 9 weeks (median 5.5 weeks, range 3–9 weeks). All control MOGAD-ON patients were specifically asked if they suffered from recent head trauma, which they had not.

4. Discussion

We describe for the first time six young patients (five children) with MOGAD-ON which may have been precipitated by head trauma, thus illustrating a novel potentially contributory etiology to the demyelinating disorder. We cannot determine if these patients were predisposed to MOGAD by a prior childhood exposure or environmental factor and the trauma acted as a “second hit” eliciting the ON or, alternatively, if the trauma was sufficient on its own to trigger MOGAD. In the former scenario, it is conceivable that there were already circulating anti-MOG antibodies or specific antibody-secreting immune cells primed to act upon an appropriate trigger (head trauma), however these children had neither a pertinent past medical history nor did they develop other stigmata of demyelination later. It is striking each of these patients paralleled one-another in history of trauma while playing sports or incidental striking by their environment, followed by a lack of symptoms until the onset of the ON. This suggests the trauma itself may be an inciting event triggering active inflammation of the optic nerve rather than a coincident inflammatory event, such as an infection, that so happened to be missed in the peri-trauma period. It is notable however,

that there were no differences in the demographic and other clinical measures between the trauma group and control group of non-trauma MOGAD-ON patients, which may challenge stratifying these patients from idiopathic causes. For example, we may have otherwise hypothesized that the trauma group was disproportionately male while the non-trauma group was disproportionately female.

While none of the patients have relapsed since the initial ON, we acknowledge it has likely not been long enough since the events to counsel families they are “out of the window.” Additionally, studies out of the Mayo group demonstrate continued asymptomatic optic nerve enhancement in a minority of patients with prior MOGAD-ON, almost universally in the same locations as the original ON (Tajfirouz et al., 2024), necessitating follow up in these patients to identify continued inflammatory insult. Additional/future episodes of mild traumatic head injuries temporally associated with relapse of ON would provide further evidence of a potential etiology of MOGAD-ON, but none of these patients have had additional head injuries to our knowledge.

Limitations of this descriptive study include the retrospective design which prevents a sophisticated evaluation of pre-trauma exposures to better understand the biologic basis of disease. We also acknowledge that given the retrospective nature of the study, recall bias may have factored into the history of trauma in our six patients. Nevertheless, given the cohort of patients with a similar history we feel that it is important to raise the possibility of trauma as a potential trigger. Further, this study relies on clinical observation with qualitative and occasionally subjective measures that cannot be standardized or controlled. Continued questions that arise from our clinical observation include: 1) is head trauma merely coincident with another inciting exposure or event to trigger MOGAD? 2) if head trauma is a true potential trigger of MOGAD, is it only a precipitant of ON or can it elicit other stigmata of MOGAD such as myelitis or cortical encephalitis? 3) if head trauma is a true potential trigger of MOGAD, is it sufficient on its own or does it require an underlying genetic or other environmental predisposition, and could it be the “first hit” of a two-hit trigger for MOGAD? and 4) what is the biologic basis of head trauma induced MOGAD? Future large-scale studies are necessary, much like those that have demonstrated no association between head trauma and the risk of multiple sclerosis in more than 110,000 trauma patients over 30 years in the UK (Goldacre et al., 2006) to elucidate the answer to these questions. Additional clinical observation from other neurology/ophthalmology groups serve as a reasonable foundation to better understand the risk of this phenomenon.

CRedit authorship contribution statement

Alexander J. Sandweiss: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Writing – original draft, Writing – review & editing. **Jonathan Rosen:** Data curation, Conceptualization, Writing – original draft, Writing – review & editing. **Chaitanya Aduru:** Conceptualization, Data curation, Writing – original draft, Writing – review & editing. **Akansha Chandrasekar:** Conceptualization, Data curation, Writing – original draft, Writing – review & editing. **Kyla Blasingame:** Conceptualization, Data curation, Writing – original draft, Writing – review & editing. **Madhuri Chilakapati:** Supervision, Writing – original draft, Writing – review & editing. **Rod Foroozan:** Supervision, Writing – original draft, Writing – review & editing. **Jonathan M. Yarimi:** Conceptualization, Data curation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

None.

Data availability

Anonymized data not published within this article will be made available by request from any qualified investigator.

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